Language-Conditioned QAOA for Portfolio Selection: A Bias-Controlled Study Reveals a Training-Horizon Confound

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Abstract. We express the possible oversight of using large language models in portfolio optimization pipelines, we include in the pipeline a language-conditioned QAOA with the LLM's role to read financial news from RavenPack and per-asset conviction that tilt the cost Hamiltonian of a cardinality-constrained Quantum Approximate Optimization Algorithm (QAOA) portfolio optimizer. With strong bias controls, we ask whether the language adds out-of-sample value to two LLMs, qwen2.5-14B, Sept. 2024 released during the evaluation window and an out of sample Llama 2, cut off Sept. 2022 on the same headlines, the qwen model beats matched random tilts by $+0.76\%$/quarter and Llama 2 -0.23%/quarter ($p = 0.94$) which is indistinguishable from random. The benefit is noticed only for the model released into the test period consistent with training-horizon hindsight rather than prediction, a misleading alpha. The pre-cutoff model is also smaller, a caveat we note. We contribute the language to quantum conditioning pipeline, a reusable bias-control template, and a quantified training-horizon look-ahead warning for LLM-for-finance evaluation.

Keywords: QAOA · Quantum optimization · LLM-conditioning · Look-ahead bias · Quantum finance.

1 Pipeline and Method

We bring attention in particular to the use case of portfolio optimization in financial institutions. A similar study has been undertaken by Chen et al. [1] and Lee et al. [2], we borrow their insights on using Black–Litterman in quantum numeric views and LLM optimized classically and question the combination of quantum and LLM in one pipeline. In this study we use a fixed set of Dow 30 applied across 2022-24, the dataset is split into 378 and 63 trading days as train and test window respectively, we perform 400 random assignments where 14 of the 30 names are made into a subset without replacement totaling 1600 unique subsets. QAOA assigns each asset as one qubit and the exact statevector simulation costs 2^{14} amplitudes, the exact optimum over $\binom{14}{5} = 2002$ portfolios. Quantum-simulation tractability is retained by limiting the sample size to 14 and drawing 1600 unbiased random subsets.

Each subset is shortlisted into a portfolio of $B = 5$ assets using the Markowitz model, μ a vector of expected returns and x the stocks held against the risk

minimized with covariance matrix Σ, later converted to a score to maximize with risk aversion $q = 2$. The base μ and Σ are estimated from CRSP total returns. We use x in binary as a combinatorial of the 2002 feasible portfolios, a candidate for a quantum optimizer.

The LLMs qwen 2.5-14B, its training plausibly spans our 2024 test window and Llama 2, pretraining cutoff Sept. 2022, before our test data window, are provided with the RavenPack (headlines tagged to companies with timestamps) and are assigned to determine whether to nudge μ given the news headline by s ($\pm 10\%$), added to the Black–Litterman expected returns $\mu \to \mu + s\,v$, shifting a stock's expected annual return. The output from the LLM is a conviction $v_i \in [-1, 1]$. Two controls, the placebo keeps the same size and sparsity but randomizes which stocks, the LLM's choice of names and direction is removed, this tests the language vs the underlying training sample of the LLM.

In the quantum layer, we hand the binary "pick 5 of 14" problem to a quantum optimizer and use QUBO to build our portfolio, given our mean variance score and "hold exactly 5" rule as the quadratic objective. We then use QAOA setting the layers or $p = 4$. Our question is whether the LLM's view changes the best portfolio at that exact optimum. Upon measurement we note at $p = 4$, QAOA's sampled portfolio equals the true optimum only 7.7% of the time, a weak exact optimizer at this depth.

2 Results

The pipeline is that two LLMs read the same headlines and go through the same simulations, qwen 2.5-14B that is more recent on changed-optimum instances, those changes beat random changes of the same size by $+0.76\%$/quarter whereas Llama 2 the same kind of changes do -0.23%/quarter. The edge appears to be apparent for the model whose training overlaps the test window. When swapped to an out of sample model holding the market and other factors the same, the edge disappears which points to one of our core findings that the edge is about the model's training horizon, not the market. Sentiment analysis comparison between qwen and ESS show qwen changes 68% of optima, +1.20%, ESS changes only 16%, +0.14%, though the ESS is window-averaged and rarely crosses the $\pm 10\%$ threshold, if ESS is matched in magnitude to qwen, at equal tilt strength, the LLM extracts stronger, more actionable views.

Limitations. The 1600 instances are subsets within 4 dates, pseudo-replicated and not a time series. Qwen is 14B parameter, Llama 2 is 7B parameter model, qwen has no published training cutoff. The limitations restrict causal attribution.

References

1. Chen, C., Chung, S., Goan, H.: Black-litterman portfolio optimization with noisy intermediate-scale quantum computers. arXiv:2312.00892 (2023)
2. Lee, Y., Kim, Y., Kim, J., Kim, S., Lee, Y.: LLM-enhanced Black-Litterman portfolio optimization. In: CIKM Workshop on Financial AI (2025)

Does Language Really Help a Quantum Portfolio Optimizer?

A Bias-Controlled Study and a Training-Horizon Confound in LLM-Conditioned QAOA

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The Central Question

When a backtest says "LLM news-views help a quantum portfolio optimizer," is that **signal — or an artifact?** We build a deliberately bias-controlled pipeline to catch ourselves.

Experimental Pipeline

01

Universe Selection

Dow 30, CRSP total returns — objective, fixed universe

02

LLM Conviction Generation

LLM reads RavenPack headlines time-stamped **strictly before** each decision date → per-asset ±10% view tilts

03

QAOA Cost Hamiltonian

Conviction tilt applied to QAOA cost function → hold 5 of 14 assets, evaluate held-out next quarter

04

Controls Applied

Matched random-tilt placebo; ablation vs. RavenPack sentiment; **two LLMs on identical headlines**

The Two-LLM Design

qwen2.5-14B

Training overlaps 2024 — **contaminated**

Llama 2

Sept-2022 cutoff — genuinely **out-of-sample**

Both models read the *same* RavenPack headlines. Only the training-horizon relationship to the backtest period differs.

⚠ **Key Finding — The apparent "language signal" is the LLM's hindsight:** The out-of-sample alpha collapses to indistinguishable-from-random with a genuinely out-of-sample model.

+0.76% per quarter

qwen2.5-14B Premium

Over random tilts (p≈0) — training overlaps 2024 backtest window

−0.23% per quarter

Llama 2 Premium

Over random tilts (p=0.94) — Sept-2022 cutoff, indistinguishable from random

47–68%

Optima Changed

A ±10% view changes the exact optimum in 47–68% of instances — language has real mechanism leverage on the selection.

7.7%

True Optimum Found

Shallow QAOA is a weak exact optimizer — no quantum-advantage claim is made

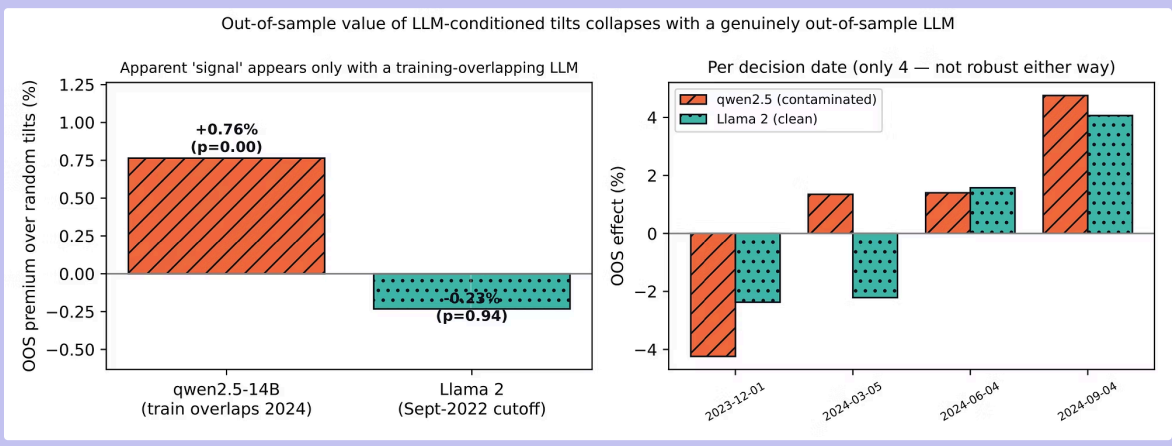


FIGURE 1 — Out-of-sample value of LLM-conditioned tilts collapses with a genuinely out-of-sample LLM.

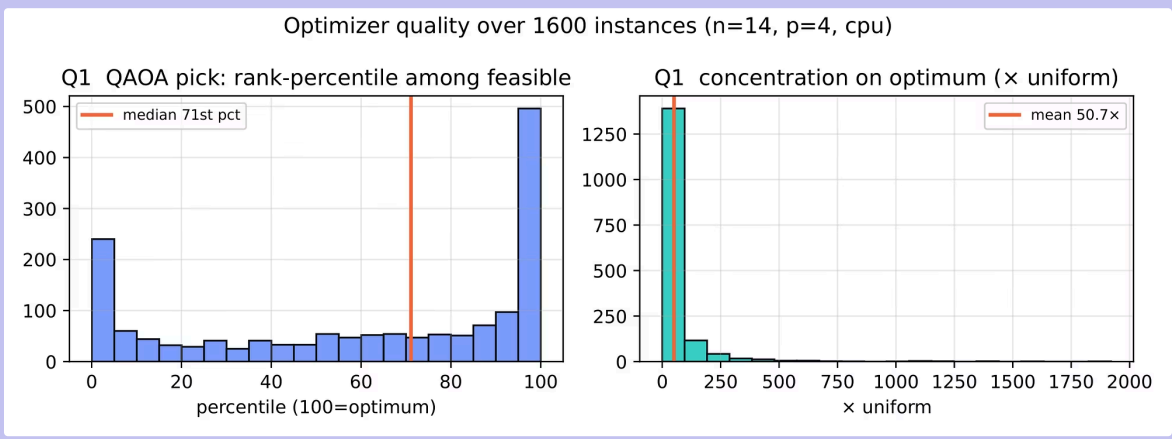


FIGURE 2 — QAOA optimizer quality over 1600 instances (n=14, p=4): rank-percentile among feasible portfolios (median 71st) and concentration on the optimum (mean 50.7× uniform).

Honest Limitations

- Only **4 decision dates** — instance-level p is pseudo-replicated
- Clean and contaminated LLMs differ in **capability** as well as cutoff date
- Small-n **exact simulation**; large-cap US equities only; no transaction costs
- None of these limitations **inflate** the central result

Contribution

→ A **reproducible bias-control template** for LLM-conditioned financial backtests

→ A **quantified warning**: date your LLM against your backtest window, or training-horizon look-ahead will manufacture alpha that isn't there

✓ **Reproducibility:** Fixed seeds, committed CRSP + conviction caches; code released upon publication.